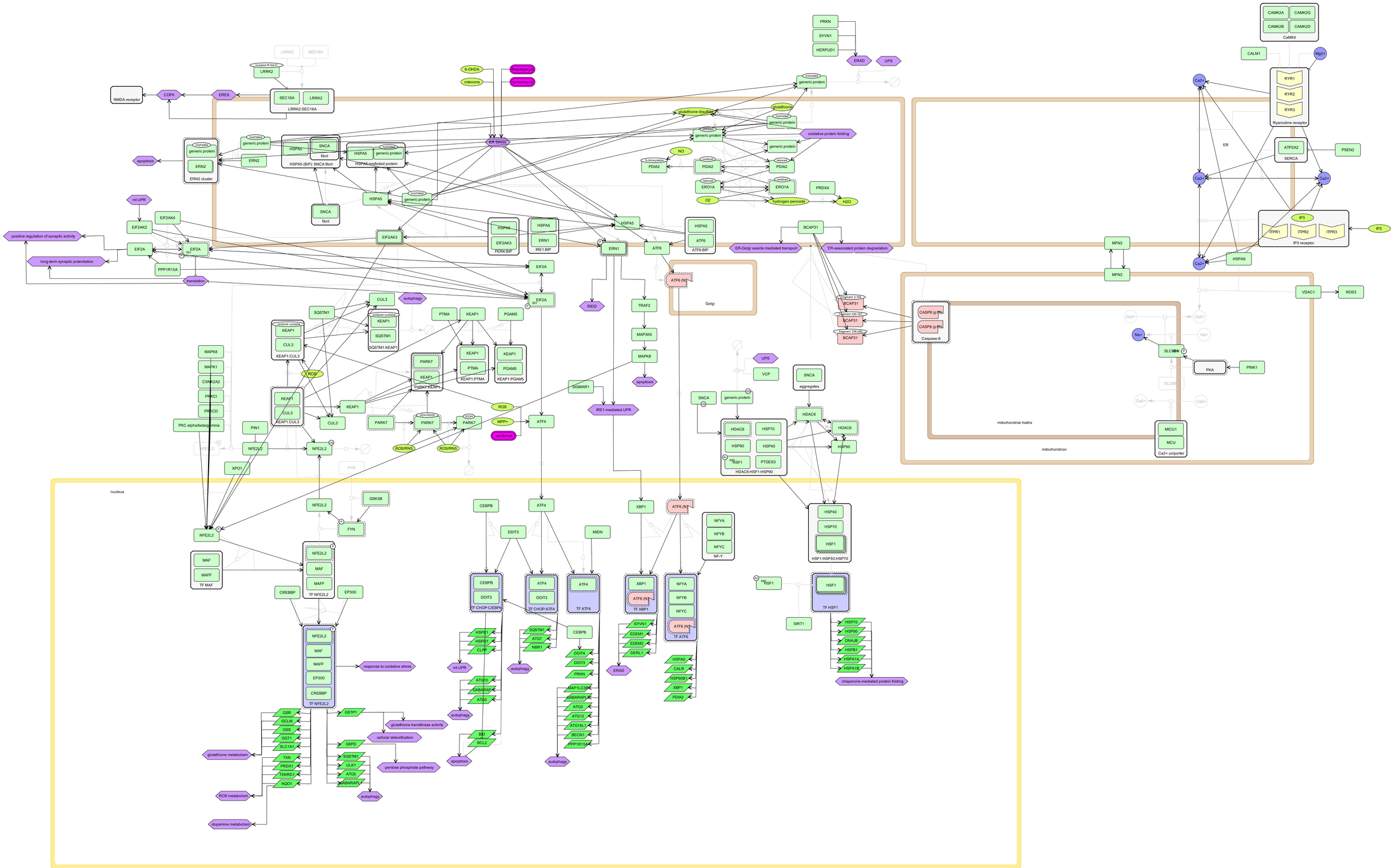

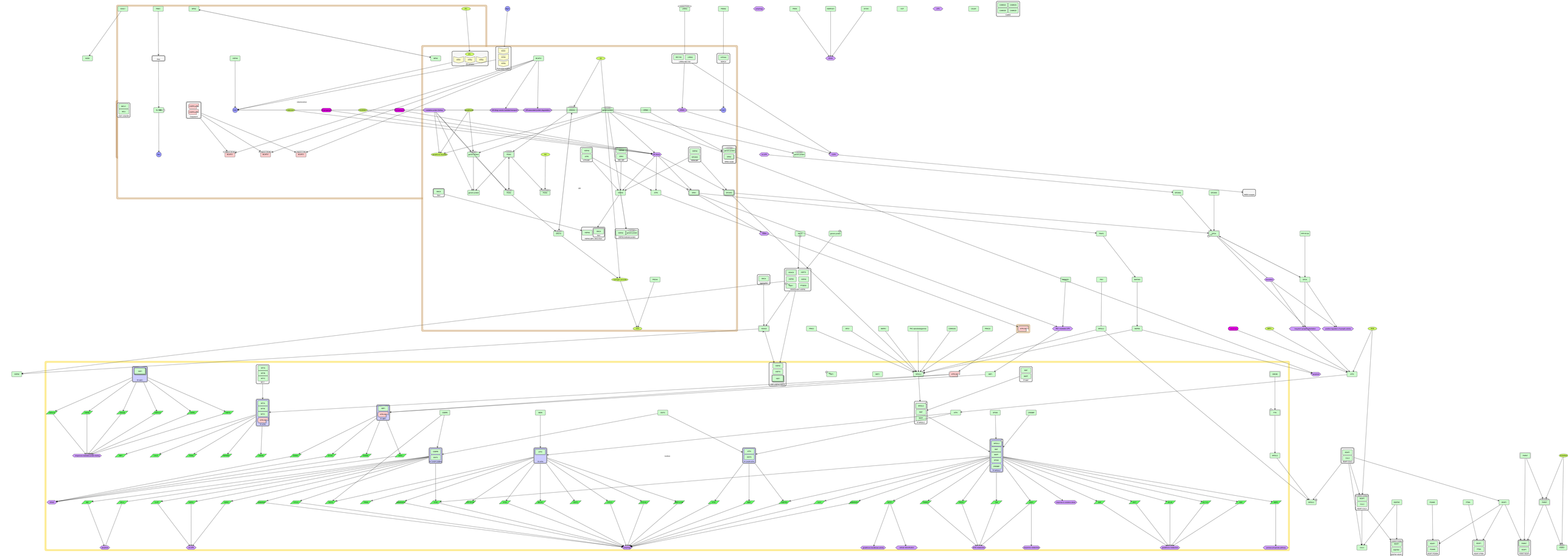

LRRK2
SEC16A
mutated R1441C
LRRK2
NMDA receptor
COPII
ERES
SEC16A
LRRK2
LRRK2:SEC16A
6-OHDA
rotenone
thapsigargin
tunicamycin
PRKN
SYVN1
HERPUD1
ERAD
UPS
misfolded
generic protein
CAMK2A
CAMK2G
CAMK2B
CAMK2D
CaMKII
CALM1
Mg2+
Ca2+
RYR1
RYR2
RYR3
Ryanodine receptor
glutathione disulfide
glutathione
misfolded
generic protein
unfolded
generic protein
oxidative protein folding
generic protein
misfolded
generic protein
misfolded
generic protein
ERN2
ERN2 cluster
apoptosis
ERN2
HSPA5
SNCA
fibril
HSPA5 (BiP): SNCA fibril
HSPA5
misfolded
generic protein
HSPA5:misfoded protein
ER Stress
NO
S-nitrosylation
PDIA2
oxidized
PDIA2
reduced
PDIA2
reduced
ERO1A
oxidized
ERO1A
PRDX4
O2
hydrogen peroxide
H2O
ER
ATP2A2
SERCA
PSEN2
Ca2+
Ca2+
mt UPR
EIF2AK2
EIF2AK4
HSPA5
misfolded
generic protein
SNCA
fibril
positive regulation of synaptic activity
EIF2A
EIF2A
S51
P
long-term synaptic potentiation
PPP1R15A
translation
EIF2AK3
HSPA5
EIF2AK3
PERK:BIP
HSPA5
ERN1
IRE1:BIP
HSPA5
P
ERN1
ATF6
HSPA5
ATF6
ATF6:BIP
ER-Golgi vesicle mediated transport
BCAP31
ER-associated protein degradation
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
IP3
MFN2
MFN2
HSPA9
Ca2+
EIF2A
EIF2A
S51
P
RIDD
ATF6 (N)
Golgi
TRAF2
fragment 2-164
BCAP31
fragment 235-167
BCAP31
fragment 238-246
BCAP31
CASP8 (p18)
CASP8 (p10)
Caspase-8
VDAC1
NOS3
Ca2+
Ca2+
Na+
Na+
SLC8B1
P
PKA
PINK1
SLC8B1
Ca2+
Ca2+
mitochondrial matrix
MICU1
MCU
Ca2+ uniporter
mitochondrion
SQSTM1
CUL3
autophagy
oxidized cysteine
KEAP1
CUL3
KEAP1:CUL3
oxidized cysteine
KEAP1
SQSTM1
SQSTM1:KEAP1
PTMA
KEAP1
PGAM5
PARK7
KEAP1
PARK7:KEAP1
KEAP1
PTMA
KEAP1:PTMA
KEAP1
PGAM5
KEAP1:PGAM5
MAP3K5
MAPK8
apoptosis
SIGMAR1
IRE1-mediated UPR
UPS
VCP
SNCA
aggregates
MAPK8
MAPK1
CSNK2A2
PRKCI
PRKCD
PKC alpha/beta/gamma
ROS
KEAP1
CUL3
KEAP1:CUL3
KEAP1
CUL3
PARK7
SOH/SO2H
PARK7
SO3H
PARK7
ROS
MPP+
salubrinal
ATF4
SNCA
Ub
generic protein
Ub
HDAC6
HSP70
HSP90
HSP40
Ac
K80
HSF1
PTGES3
HDAC6:HSF1:HSP90
HDAC6
HDAC6
HSP90
PIN1
NFE2L2
NFE2L2
Ub
NFE2L2
XPO1
ROS/RNS
ROS/RNS
FYN
nucleus
NFE2L2
GSK3B
FYN
P
NFE2L2
P
MAF
MAFF
TF MAF
NFE2L2
P
MAF
MAFF
TF NFE2L2
CREBBP
EP300
NFE2L2
P
MAF
MAFF
EP300
CREBBP
TF NFE2L2
response to oxidative stress
CEBPB
ATF4
DDIT3
MIDN
XBP1
ATF6 (N)
NFYA
NFYB
NFYC
NF-Y
HSP40
HSP70
HSF1
HSF1:HSP40:HSP70
CEBPB
DDIT3
TF CHOP:C/EBPb
ATF4
DDIT3
TF CHOP:ATF4
ATF4
TF ATF4
XBP1
ATF6 (N)
TF XBP1
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
Ac
K80
HSF1
HSF1
TF HSF1
SIRT1
CEBPB
HSPE1
HSPD1
CLPP
mt UPR
SQSTM1
ATG7
NBR1
autophagy
DDIT4
DDIT3
PRKN
SYVN1
EDEM1
EDEM2
DERL1
ERAD
HSPA5
CALR
HSP90B1
XBP1
PDIA2
HSP70
HSP90
DNAJB
HSPB1
HSPA1A
HSPA1B
chaperone-mediated protein folding
ATG10
GABARAP
ATG5
autophagy
MAP1LC3B
GABARAPL2
ATG3
ATG12
ATG16L1
BECN1
PPP1R15A
autophagy
BID
BCL2
apoptosis
GSR
GCLM
GSS
GGT1
SLC1A1
glutathione metabolism
TXN
PRDX1
TXNRD1
NQO1
ROS metabolism
dopamine metabolism
GSTP1
glutathione transferase activity
cellular detoxification
G6PD
pentose phosphate pathway
SQSTM1
ULK1
ATG5
GABARAPL1
autophagy

VDAC1
PINK1
MFN2
IP3
Mg2+
NOS3
PKA
HSPA9
MFN2
IP3
ITPR1
ITPR2
ITPR3
IP3 receptor
RYR1
RYR2
RYR3
Ryanodine receptor
BCAP31
mitochondrion
MICU1
MCU
Ca2+ uniporter
SLC8B1
P
CASP8 (p18)
CASP8 (p10)
Caspase-8
Ca2+
rotenone
thapsigargin
6-OHDA
tunicamycin
oxidative protein folding
glutathione
ER-Golgi vesicle mediated transport
ER-associated protein degradation
Na+
fragment 235-167
BCAP31
fragment 238-246
BCAP31
fragment 2-164
BCAP31
glutathione disulfide
unfolded
generic protein
oxidized
PDIA2
NO
SNCA
fibril
generic protein
reduced
PDIA2
S-nitrosylation
PDIA2
mutated R1441C
LRRK2
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CAMK2A
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CAMK2B
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SEC16A
LRRK2
LRRK2:SEC16A
ATP2A2
SERCA
oxidized
ERO1A
misfolded
generic protein
ERN2
ERES
Ca2+
HSPA5
ATF6
ATF6:BIP
HSPA5
ERN1
IRE1:BIP
ER Stress
HSPA5
EIF2AK3
PERK:BIP
misfolded
generic protein
ERN2
ERN2 cluster
mt UPR
misfolded
generic protein
COPII
ER
HSPA5
ATF6
P
ERN1
EIF2AK3
reduced
ERO1A
HSPA5
SNCA
fibril
HSPA5 (BiP): SNCA fibril
HSPA5
misfolded
generic protein
HSPA5:misfoded protein
RIDD
SNCA
Ub
generic protein
Ub
hydrogen peroxide
PRDX4
SNCA
aggregates
HDAC6
HSP70
HSP90
HSP40
Ac
K80
HSF1
PTGES3
HDAC6:HSF1:HSP90
SIGMAR1
EIF2AK2
EIF2AK4
NMDA receptor
TRAF2
EIF2A
P
S51
PPP1R15A
H2O
HDAC6
PRKCI
XPO1
MAPK1
PKC alpha/beta/gamma
CSNK2A2
PRKCD
ATF6 (N)
IRE1-mediated UPR
SIGMAR1
PIN1
MAP3K5
NFE2L2
MAPK8
translation
EIF2A
salubrinal
MPP+
long-term synaptic potentiation
positive regulation of synaptic activity
ROS
HSP90
HSF1
TF HSF1
NFYA
NFYB
NFYC
NF-Y
NFYA
NFYB
NFYC
ATF6 (N)
TF ATF6
XBP1
ATF6 (N)
TF XBP1
CEBPB
MIDN
HSP40
HSP70
HSF1
HSF1:HSP40:HSP70
Ac
K80
HSF1
SIRT1
NFE2L2
P
ATF6 (N)
XBP1
MAF
MAFF
TF MAF
GSK3B
apoptosis
ATF4
NFE2L2
MAF
MAFF
TF NFE2L2
MIDN
DDIT3
ATF4
EP300
CREBBP
P
FYN
HSPA1B
HSP90
DNAJB
HSPA1A
HSPB1
HSP70
chaperone-mediated protein folding
XBP1
CALR
PDIA2
HSP90B1
HSPA5
EDEM1
SYVN1
EDEM2
DERL1
CEBPB
DDIT3
TF CHOP:C/EBPb
ATF4
TF ATF4
nucleus
ATF4
TF ATF4
ATF4
DDIT3
TF CHOP:ATF4
NFE2L2
MAF
MAFF
EP300
CREBBP
TF NFE2L2
NFE2L2
KEAP1
CUL3
KEAP1:CUL3
PARK7
ROS/RNS
ERAD
BID
BCL2
CLPP
HSPD1
HSPE1
GABARAP
ATG10
DDIT4
PRKN
GABARAPL2
ATG5
PPP1R15A
ATG3
BECN1
DDIT3
ATG12
ATG16L1
MAP1LC3B
ATG7
NBR1
SQSTM1
ULK1
GABARAPL1
GSTP1
TXNRD1
PRDX1
TXN
NQO1
response to oxidative stress
GSR
GGT1
GCLM
SLC1A1
GSS
G6PD
NFE2L2
Ub
oxidized cysteine
KEAP1
CUL3
KEAP1:CUL3
SQSTM1
PGAM5
PTMA
KEAP1
SOH/SO2H
PARK7
apoptosis
mt UPR
autophagy
glutathione transferase activity
cellular detoxification
ROS metabolism
dopamine metabolism
glutathione metabolism
pentose phosphate pathway
CUL3
oxidized cysteine
KEAP1
SQSTM1
SQSTM1:KEAP1
KEAP1
PGAM5
KEAP1:PGAM5
KEAP1
PTMA
KEAP1:PTMA
PARK7
KEAP1
PARK7:KEAP1
SO3H
PARK7